Late Payment by Large UK Companies: A Statistical Analysis

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Overview

I analysed the reports in which large UK companies publish how quickly they pay their suppliers. I chose them because late payment can harm the cash flow and trading of the businesses waiting to be paid, especially smaller ones (Department for Business, Innovation, Science and Trade [BIST], 2025, Introduction section). The dataset is the GOV.UK payment practices reports ("Check when large businesses pay their suppliers"), at https://www.gov.uk/check-when-businesses-pay-invoices (BIST, 2026). My question was: is the share of invoices paid late lower in 2023 to 2024 than in 2018 to 2019, for the same large UK companies?

Dataset Description

Each row is one large company's statutory report on how it paid its suppliers over a reporting period, and a company normally has two reporting periods in each financial year (Reporting on Payment Practices and Performance Regulations 2017, reg. 7). The export I took on 6 October 2026 loads as 115,358 rows and 52 columns, and dropping the 10 free-text columns (text columns whose median answer is longer than eight words) at load leaves 42. The outcome is the share of invoices paid late, meaning the data column "% Invoices not paid within agreed terms," a numeric figure that suits both descriptive statistics and a paired test. I also use the average time to pay, the three time bands (within 30 days, 31 to 60 days and later than 60 days), the period end and filing dates, and as grouping variables the company number, Policy Regime and the payment code answer. Policy Regime seems to label the version of the reporting rules a report was filed under, and I explain how I infer its three values under Harmonisation to Regime-

1. GOV.UK now gives the Department for Business and Trade (DBT) a new name, the Department for Business, Innovation, Science and Trade (DBT, n.d.), so I credit it with the guidance and the data, published under the Open Government Licence v3.0 (BIST, 2026).

Methods

Initial Data Analysis and Screening

I followed initial data analysis, which "commonly takes place between the end of data retrieval and the beginning of data analysis that addresses the research question" (Lusa et al., 2024, p. 2). Two of its six steps are data cleaning and data screening (Lusa et al., 2024, p. 2), and my anomaly list covers both. Before I ran the screening in the notebook, I fixed the question, the hypothesis test and its significance level (α), because the article asks for a defined question and a known analysis strategy first (Lusa et al., 2024, p. 4).

I loaded and described every column, then listed every anomaly I tested for with one function, and added one more check for figures that cannot all be true. Two checks found nothing: no report ends before it starts, and no percentage lies outside 0 to 100. More than one report from a company in a period is normal, and one mean per company per period deals with it. Every other anomaly got one ruling (Table 1). Each ruling marks its reports with a flag and a flag_reason, and no raw row is deleted. A change to the plan after screening is allowed, but it "should be thoroughly documented and justified to ensure complete transparency" (Lusa et al., 2024, p. 27).

Table 1

The Ruling on Each Anomaly

Anomaly	Reports	Ruling
Average time to pay above 365 days (largest 4,323,394 days)	49	Excluded from the fitted model only
Average time to pay above an assumed 120-day limit, up to 365 days	309	Kept
Start or end date outside 2017 to 2026	34	Excluded
All three time bands are 0	178	Kept, left out of the band-based checks
Superseded report (1,861 company periods filed more than once, 3,782 rows)	1,921	Excluded, the latest-filed report stands
Share of invoices paid late is blank	9,423	Excluded, no outcome to measure
Regime-1 report with a dispute percentage, a figure the 2017 rules did not require	87,188	Kept, the dispute percentage is not used
Figures that cannot all be true	370	Excluded, results shown with and without
Report filed before its period ends (the furthest by 365,207 days)	88	Kept, late publication counts them as on time

Note. The 120-day, 365-day and ten-year limits are my own assumptions, set before the build, not limits from the publisher or the law. The data's 99th percentile for average time to pay is 91 days.

For a period filed more than once, I keep the latest-filed report, as the government's own statistics do (DBT, 2026, p. 20). Their July 2026 methodology still carries the department's former name. The same statistics also remove reports whose periods overlap by more than a month (DBT, 2026, p. 19). In this export, 503 reports overlap an earlier one from the same company by more than 30 days, 227 of them among the reports the analysis uses. I kept them, because the mean change is -5.20 percentage points with them and without them. The 88 reports

filed before their period ended may carry a wrong date, or the company may have published early, so I flag them and keep them. All 9,423 blank outcomes come from reports with no qualifying contracts, so they are empty reports, not missing answers. Figures that cannot all be true give an average time to pay outside what their own time bands allow, even after rounding. Of the 370, 355 give an average below what their bands allow. The other 15 give one above it, and for those I also assume no single payment takes more than ten years. The four excluding rulings make up the excluded rows, 10,271 reports (Table 2).

Table 2

Counts With and Without the Excluded Rows

Count	Every report	Without the excluded rows
Reports	115,358	105,087
Regime-1 reports	99,861	90,792
Regime-1 reports ending in the two periods	54,759	49,512
Companies in 2018 to 2019	7,476	7,468
Companies in 2023 to 2024	6,779	6,774
Paired companies	5,376	5,371

Harmonisation to Regime-1

The export labels each report Regime-1, Regime-2 or Regime-3, and I found no definition of the labels on the publisher's pages. From the dates and the legislation, I infer that Regime-1 reports were filed under the original 2017 rules. The other two follow amending statutory instruments (SI) whose reporting changes apply from 2025. They are the Reporting on Payment Practices and Performance (Amendment) Regulations 2024 and the Reporting on Payment Practices and Performance (Amendment) Regulations 2025. The first, SI 2024/444, changed when a payment counts as made where a finance provider is involved (Reporting on Payment Practices and Performance (Amendment) Regulations 2024, reg. 2(4)(d)). Initial data

analysis warns that a lack of harmonised variables can bring "inconsistent definitions" and "information bias" (Lusa et al., 2024, p. 26). So the analysis uses Regime-1 reports only, all measured under one definition.

The Analysis Table

I use "company" for every business that reports, including limited liability partnerships, which report under parallel regulations (BIST, 2025, Introduction section). I place each report in a period by the year its reporting period ends, grouping the years into two intervals, which initial data analysis suggests for describing data measured at varying times (Lusa et al., 2024, p. 5). The analysis table gives each of the 5,371 paired companies, the companies that reported in both periods, one unweighted mean share of invoices paid late per period. Across every company with a figure in a period, most figures average four reports, two a year for two years. That holds for 4,526 of the 7,468 companies in 2018 to 2019 and 5,132 of the 6,774 in 2023 to 2024. The rest average between one and ten.

Participation and Drop-Out

Because the test can use only the paired companies, I sorted every company with a figure in either period into stayers (the paired companies), drop-outs (a figure in 2018 to 2019 only) and new entrants (a figure in 2023 to 2024 only). The article asks analysts to "Describe the participants who dropped out from the study during the follow-up" (Lusa et al., 2024, p. 6). I describe each group without a test.

Why These Descriptive Statistics

For each numeric column I report the mean, standard deviation, median and quartiles, with the skew of the five variables the analysis rests on (average time to pay, the three time bands and the share of invoices paid late). For each grouping column I give the count of each

value, and for payment codes "No" against any other answer. Screening the distributions, as initial data analysis recommends (Lusa et al., 2024, p. 8), showed a few extreme values, so I describe average time to pay by its median. That choice rests on my own figures: a median of 33 days against a mean of 112.91. For the share of invoices paid late I give both, because the test works on means and the median shows the typical report.

Why Each Type of Model

The brief defines a visual model as a graph or chart. So this project has three visual models and one fitted statistical model, and no machine learning model. The brief says I am not training machine learning models yet and limits the tools to Python, NumPy, pandas, Matplotlib, Seaborn, SciPy and Jupyter, so I meet the rubric's deep learning rows with the fitted model and the rubric's own headings.

- **Visual model 1**, the median share of invoices paid late by year, shows the trend in the typical report over time, which initial data analysis asks for (Lusa et al., 2024, p. 7).
- **Visual model 2**, a boxplot of the paired companies in each period, compares the two periods' centre and spread at a glance.
- **Visual model 3**, a scatter of each paired company's two figures, keeps each pair together, as the test does. It does the job of a profile plot, which shows how the outcome changes and varies within each subject (Lusa et al., 2024, p. 7), for two periods.
- **The fitted model** is a least-squares line of the share of invoices paid late against the average time to pay, the plainest other measure of how a company pays. I trained it on reports ending in 2018 to 2022 and evaluated it on reports ending in 2023 to 2024, which it never saw. What matters is how accurately a method predicts data it has not seen (James et al., 2023, p. 28). I judge its fit by R^2, the proportion of the variation in the

outcome that the line explains (James et al., 2023, p. 79). A held-out set is often drawn at random (James et al., 2023, p. 202), but splitting by time also tests whether the line holds in later years.

Why the Paired *t*-Test Is Appropriate

The question compares the same companies in two periods. Each paired company's two means are strongly related (Pearson's $r = .64$). Initial data analysis expects this: "The repeated measurements from the same subject in longitudinal studies are usually correlated" (Lusa et al., 2024, p. 9). So I test each company's own change with a paired t-test. The standard deviation of the changes, 18.63 points, is smaller than the spread in either period (22.36 and 21.29), so pairing gives a more precise test. With 5,371 pairs, a t-test suits even a skewed outcome. As Fagerland (2012) puts it, "For studies with a large sample size, t-tests and their corresponding confidence intervals can and should be used even for heavily skewed data" (p. 1). Fagerland (2012, p. 1) adds that in a large study a non-parametric test may answer the wrong question, and the mean change answers mine directly. Fagerland (2012) compares two independent groups, so I cite him for the large-sample argument only. I add Cohen's d_z, the standardised mean difference for within-subjects designs (Lakens, 2013, p. 4), which is the mean change divided by the standard deviation of the changes. Effect sizes like this let researchers show the practical significance of a result (Lakens, 2013, p. 1), so I give it beside the p-value.

Assumptions for the Test

The test assumes the companies' changes are independent of each other, and that their mean is close to normal. The central limit theorem makes the second more likely as the sample grows (Fagerland, 2012, pp. 1-2). Each company counts once, however many reports it filed. I checked normality descriptively, with the skewness of the changes and a histogram and quantile-

quantile (Q-Q) plot, not with a second formal test. In large studies such tests are too sensitive to help choose a test (Fagerland, 2012, p. 6). I fixed $\alpha = .05$, two-sided, before the test ran. The p-value rests on these assumptions. It is "the probability that the chosen test statistic would have been at least as large as its observed value if every model assumption were correct, including the test hypothesis" (Greenland et al., 2016, p. 339).

The Screening Rules

I also tested every report, in all three regimes, against two rules that apply to it: the standard of a payment code the company names, and the legal deadline for publishing. The first is the code standard. In 2018 the Prompt Payment Code said "Paying invoices within 60 days is a requirement (this is met by paying 95% of invoices within this period), unless there are exceptional circumstances" (Chartered Institute of Credit Management, 2018b, "Code Signatories Undertake To" section). Its frequently asked questions page applied this to "undisputed invoices" (Chartered Institute of Credit Management, 2018a, "60 and 30 Day Payment Term" section). The Fair Payment Code has since replaced the Prompt Payment Code (Office of the Small Business Commissioner, n.d.-a, para. 8). Its Bronze Award requires firms "To pay at least 95% of all invoices within 60 days" (Office of the Small Business Commissioner, n.d.-b, Code Criteria section). A signatory report is a report that names the Prompt Payment Code or the Fair Payment Code, with hedged and indirect entries left out. The code standard is 95% of invoices paid within 60 days, the standard both codes set. A breach of the code standard is a signatory report whose own figures show fewer than 95% paid within 60 days, after allowing for rounding.

The second is the filing period: "30 days beginning with the day after the last day of the reporting period to which a report relates" (Reporting on Payment Practices and Performance

Regulations 2017, reg. 2). A company must publish "within the filing period" (Reporting on Payment Practices and Performance Regulations 2017, reg. 3(2)(a)). If it does not, the company and every person who was then a director "commits an offence" (Reporting on Payment Practices and Performance Regulations 2017, reg. 8(1)). A report is published late when it was filed more than 30 days after its reporting period ends. Both checks give counts and rates only, with no test.

Results

Descriptive Statistics

Across the 105,935 reports that give it, the share of invoices paid late has a mean of 26.22% and a median of 20%. A quarter of reports are at 39% or more. Its skew of 1.06 shows a long right tail: most reports pay most invoices on time, but some pay many late. Average time to pay has a median of 33 days but a mean of 112.91, because the 49 reports above 365 days lift the mean from 35.8 days on their own. Of the reports, 104,362 answer No on payment codes and 10,996 give any other answer. Of the 98,937 outcome values from 1 to 99, 19.18% end in 0 or 5, against 19.19% if every value were equally common. So I found none of the rounding to 5s and 10s that the article's case study found (Lusa et al., 2024, p. 20).

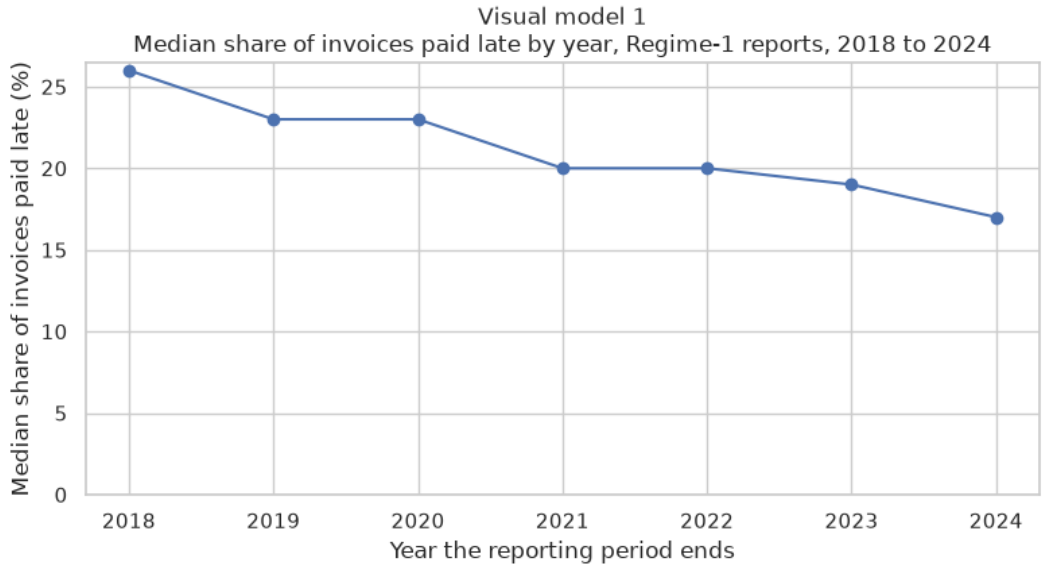
Stayers, drop-outs and new entrants look alike in the period they share. In 2018 to 2019 the 5,371 stayers had a mean share of invoices paid late of 29.49% (95% confidence interval 28.89 to 30.09). The 2,097 drop-outs had 30.14% (29.07 to 31.20), with a median of 25% in both groups. In 2023 to 2024 the 1,403 new entrants had a mean of 23.49%, against 24.29% for the stayers. Of the drop-outs, 1,917 (91.42%) filed nothing for a period ending after 2022.

The Visual Models

Visual model 1 (Figure 1) shows the typical Regime-1 report's share of invoices paid late falling from 26% for periods ending in 2018 to 17% in 2024, with no year going back up. But the companies behind each point change from year to year.

Figure 1

Median Share of Invoices Paid Late by Year, Regime-1 Reports, 2018 to 2024

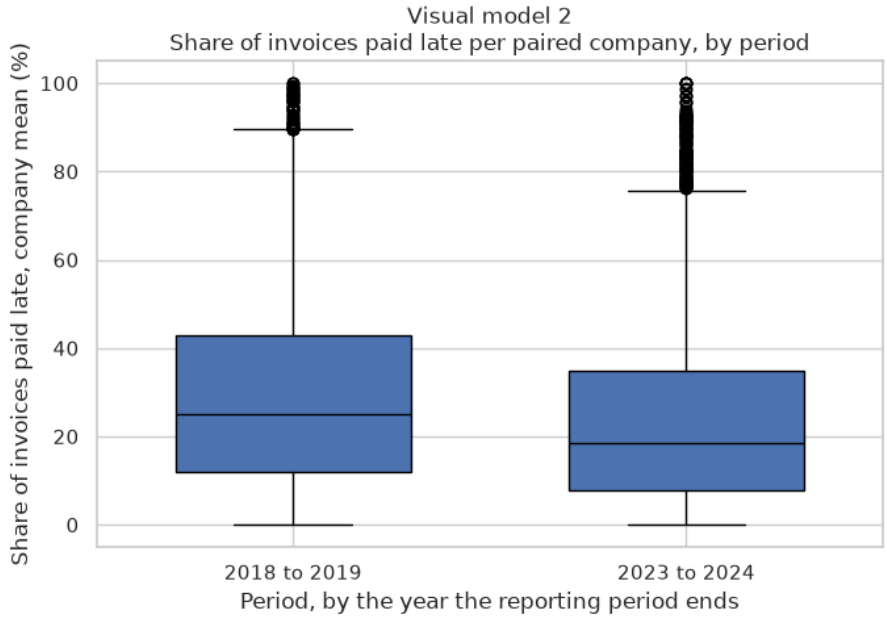


Note. Data from BIST (2026).

Visual model 2 (Figure 2) shows the whole box for the paired companies moving down. The median falls from 25% to 18.50%, and the quartiles from 12% and 43% to 7.75% and 35%. It cannot show the pairing, because a company's place in one box is not linked to its place in the other.

Figure 2

Share of Invoices Paid Late per Paired Company, by Period

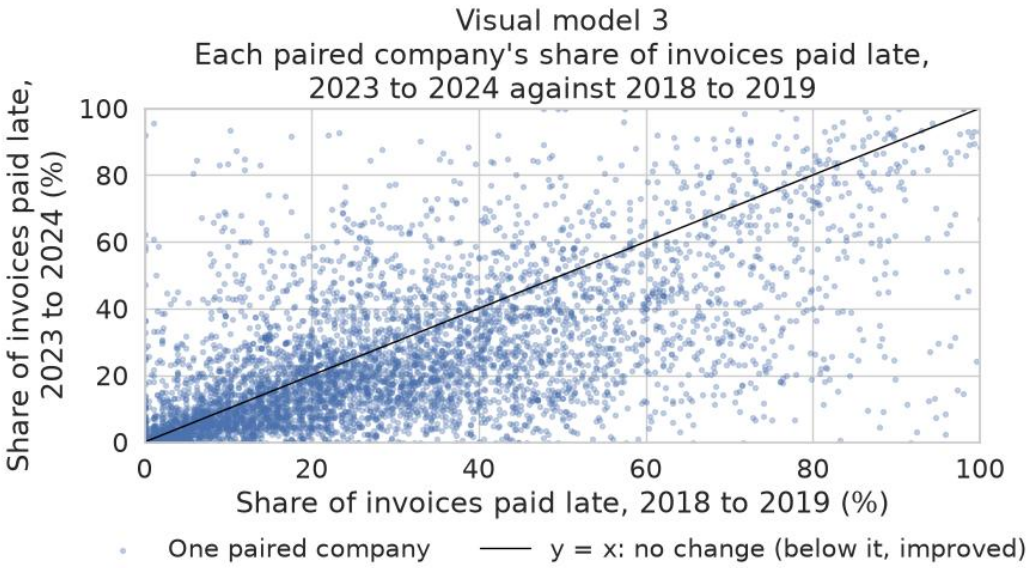


Note. Data from BIST (2026).

In Visual model 3 (Figure 3), 3,368 of the 5,371 paired companies (62.71%) sit below the line of no change, 1,834 above it and 169 on it. The points far from the line show that some companies changed a great deal in either direction.

Figure 3

Each Paired Company's Share of Invoices Paid Late, 2023 to 2024 Against 2018 to 2019



Note. Data from BIST (2026). The diagonal line marks no change.

All three point the same way, and each has a trade-off. Visual model 1 shows every year of the trend, at the cost of mixing different companies. Visual model 2 compares the two periods' spread, but drops the pairing. Visual model 3 shows the pairing directly, but shows direction better than size. I judge that Visual model 3 best supports the question, because it keeps each company's two figures together, as the question and the test do. Its weakness is size, which the test measures.

The Fitted Model

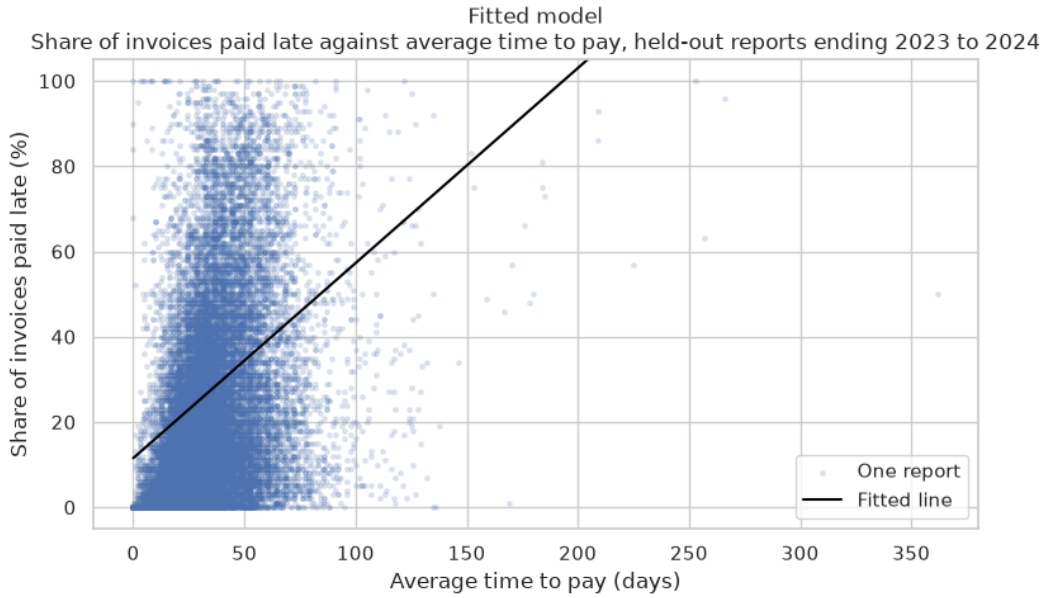
On 62,186 training reports, the fitted model's slope is 0.4575 percentage points per extra day of average time to pay, with an intercept of 11.53%. Its R^2 is .13, so the line accounts for only about 13% of the variation between reports. The analytic 95% confidence interval for the slope, 0.4482 to 0.4668, treats every report as independent. But 7,894 of the 8,334 training companies (94.72%) filed more than one. The bootstrap measures an estimate's uncertainty by

resampling the data (James et al., 2023, pp. 212-213), and I resampled whole companies to keep each company's reports together. Resampling them 2,000 times gives a bootstrap interval of 0.4303 to 0.4865, 3.0 times as wide. The slope stays well above zero either way.

On the 24,097 held-out reports, the line misses a report's share of invoices paid late by 17.00 percentage points on average, against 17.65 in training. So it does no worse on later years. But a typical miss of 17 points is large next to yearly medians of 17% to 26%. Figure 4 shows why: reports with the same average time to pay run from 0% to 100% late, and the line passes 100% at 193 days. So I read the fitted model as evidence that slower payment goes with a larger share of invoices paid late, not as a way to predict one company.

Figure 4

Share of Invoices Paid Late Against Average Time to Pay, Held-Out Reports Ending 2023 to 2024



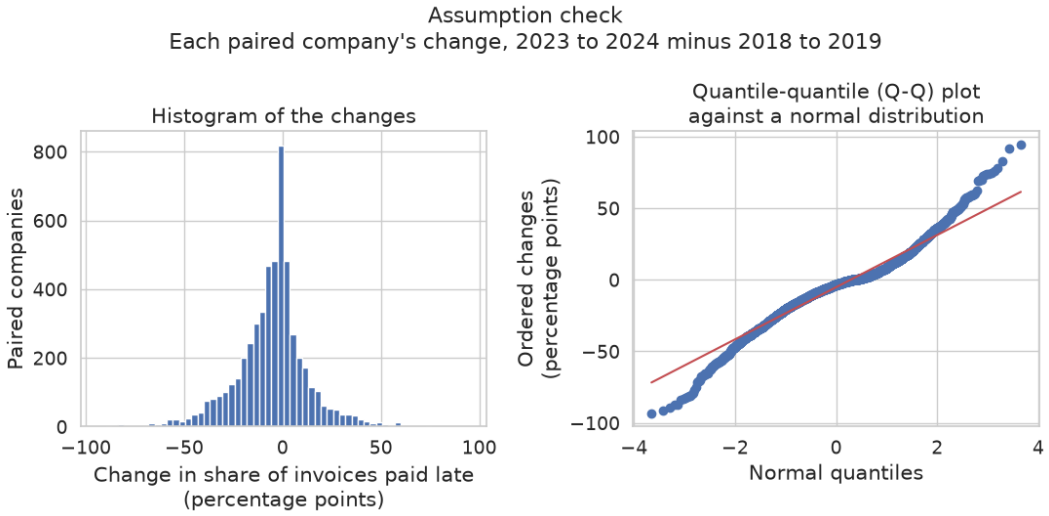
Note. Data from BIST (2026). The line was fitted on reports ending in 2018 to 2022.

The Hypothesis Test

The changes of the 5,371 paired companies are close to symmetric, with a skewness of -0.07 (Figure 5). The Q-Q plot bends away at both ends, so the tails are heavier than normal. With this many pairs and so little skew, the mean of the changes is close to normal even so.

Figure 5

Each Paired Company's Change, 2023 to 2024 Minus 2018 to 2019



Note. Data from BIST (2026). Left, a histogram of the changes. Right, a Q-Q plot against a normal distribution.

The share of invoices paid late fell by 5.20 percentage points on average, with a 95% confidence interval for the change of -5.70 to -4.70. The test gives $t(5370) = -20.45$, $p < .001$, far below $\alpha = .05$, so I reject the null hypothesis that the mean change is zero. Cohen's d_z is -0.28, so the fall is clear but modest next to how much companies differ in their change. If the test's assumptions hold and I repeated the study many times, about 95% of such intervals would contain the true mean change (Greenland et al., 2016, p. 340).

Screening Against the Reports' Own Rules

Of the 9,322 signatory reports, 8,159 from 939 companies have figures I can test. The other 1,163 have blank time bands or all three at 0. On their own figures, 2,587 of the 8,159 (31.71%) are in breach of the code standard, and 606 of the 939 companies (64.54%) published at least one such report. These two figures count different things. The first is per report and the second per company, and each company filed 8.69 tested reports on average, so one breach in any of them is enough to count. The 219 companies (23.32%) that breached in five or more reports hold 1,751 of the 2,587 breaching reports (67.68%), so most breaches come from companies that miss the standard again and again. Three caveats sit beside these breach figures. The Prompt Payment Code's bar applied to undisputed invoices and allowed exceptions, while the bands count every invoice. Membership is self-declared, and I cannot check it against the code's own list. And the figures come from each company's own report, not an audit. Still, 530 signatory reports paid 20% or more of their invoices later than 60 days, so a breach is not a matter of rounding. The share of tested signatory reports in breach falls from 58.32% for periods ending in 2018 to 22.60% in 2024.

By the regulation's own wording, 33,031 of the 115,358 reports (28.63%) were published late, and 8,010 of the 10,262 companies (78.05%) published at least one late. As with breaches, one late report among a company's many is enough to count it. As a sensitivity check, I allowed one extra day: even then, 24,560 reports (21.29%) were late. The late rate did not improve, from 29.30% for periods ending in 2018 to 30.73% in 2024. A late report may also be a correction whose on-time original was deleted (BIST, 2025, Incorrect reports section). Superseded reports count here too: 753 of the late reports are superseded reports, and without all 1,921 superseded reports, 28.45% of the rest were late.

Results With and Without the Screened Reports

The result does not rest on the excluded rows, the reports in breach of the code standard or the impossible-figures reports (Table 3). All four 95% confidence intervals lie between -5.70 and -4.39, so every run shows the same clear fall.

Table 3

The Hypothesis Test With and Without the Screened Reports

Run	Paired companies	Mean change (percentage points)	95% confidence interval
The hypothesis test, every ruling applied	5,371	-5.20	-5.70 to -4.70
Every report	5,376	-5.17	-5.67 to -4.67
Every report, without the reports in breach of the code standard	5,236	-4.90	-5.40 to -4.39
Every report, without the impossible-figures reports	5,371	-5.19	-5.69 to -4.69

Two Findings Made After Looking at the Data

I found these two comparisons after looking at the data. They are not part of the planned analysis, so I read each only as an association. First, 2,985 paired companies published at least one 2023 to 2024 report late. Their share of invoices paid late fell by 4.00 percentage points (95% confidence interval -4.70 to -3.30). For the 2,386 that published every one on time, it fell by 6.70 (-7.40 to -6.00). Second, the median tested signatory report gives 12% of invoices paid late, against 21% for the 95,971 reports that name no code and have time bands I can test.

Interpretation for a Non-Technical Audience

I looked at the same 5,371 large companies at two points, about five years apart. In 2018 to 2019, the average company paid about 29 in every 100 invoices late. By 2023 to 2024, that was down to about 24 in every 100. My test shows that if nothing had really changed, a fall this

large would almost never happen. But it is a modest change, and companies differ a lot. About six in ten companies improved, and about one in three got worse.

Two other things stood out. About a third of the tested signatory reports, those that name the Prompt Payment Code or the Fair Payment Code, show the company missing that code's standard of paying 95 in every 100 invoices within 60 days, on its own figures. Three cautions apply: the older code's bar covered only undisputed invoices and allowed exceptions, companies declare their own membership, and nobody audits the figures. Even so, in a pattern I noticed only after looking at the data, their reports show a smaller share of invoices paid late than reports that name no code. And more than one report in four was published after the legal deadline, with no improvement over the years.

Companies that take longer to pay, on average, also report a larger share of invoices paid late. That link is loose, so it tells us about companies in general, not about any one company. None of this says why late payment fell, and every figure is what the companies said about themselves.

Limitations and Potential Bias

A Dataset Limitation: The Figures Are Self-Reported

Every figure comes from the company's own report, and nobody independently verifies or audits the data (DBT, 2026, p. 6). The figures that cannot all be true show that some reports are wrong. So the fall and the breaches are what companies said.

A Source of Bias: Who Reports, and Who Stays

The department's statistics methodology notes that not every business that must report does so (DBT, 2026, p. 12). Initial data analysis would compare the participants with the target population (Lusa et al., 2024, p. 6), but the data holds no list of the companies that should report.

The test also uses only the companies that reported in both periods. Drop-out that the analysis does not take into account could bias the results (Lusa et al., 2024, p. 26). The drop-outs looked like the stayers in 2018 to 2019, but the data does not say why a company stopped. So the result describes companies that kept reporting.

Period Effects and Causes

In a longitudinal design, the effects of age, cohort and period can be hard to separate and can be confounded (Lusa et al., 2024, p. 9). Everything that changed for all companies between the two periods sits inside the mean change, with no comparison group. Possible causes include the reporting duty itself, the payment codes, and changes in trading conditions between the two periods, including the COVID-19 pandemic, and I tested none of them.

Other Limits

Each company counts once, whether its figure averages one report or ten, so the result describes the average large company, not the average invoice. The data has no invoice counts to weight by. The regime mapping is my inference. The two findings made after looking at the data were not planned, so they need a fresh test on new data. The fitted model's analytic interval was too narrow, because a company's reports move together. Initial data analysis warns that ignoring such serial correlation could bias estimation (Lusa et al., 2024, p. 27). The bootstrap corrects the interval's width only. And the export has no column linking companies in the same group, so I cannot check that their changes are independent.

An Ethical Concern: Naming Companies on Unaudited Figures

These counts could be misused to name companies as code breakers or offenders. Publishing late is an offence (Reporting on Payment Practices and Performance Regulations 2017, reg. 8). The regulations now run to 6 April 2031 (Reporting on Payment Practices and

Performance (Amendment) Regulations 2024, reg. 2(2)), so naming a company from these counts would accuse it of a crime. Yet the figures are unaudited, and a late report may be a correction, because a published report cannot be amended (BIST, 2025, Incorrect reports section). So I name no company in these counts and give counts and rates only. Anyone judging one company should read its own reports and ask it first.

An Ethical Concern: Personal Data in Free Text

Information about a limited company is not personal data (Information Commissioner's Office, n.d., What about information about companies? section). But "A name and a corporate email address clearly relates to a particular individual and is therefore personal data" (Information Commissioner's Office, n.d., What about information about companies? section). My check of the export before the build found contact email addresses in some of the free-text columns, and some may name individuals. So I drop them at load and never print one. The brief asks for the original dataset file, so the export goes in the submission zip, but I keep it out of the public repository.

Next Steps for These Limitations

Each step below answers a limitation above.

- For self-reporting, I would check a sample of breach reports against the company's own payment policy, and ask the code's administrator for its signatory list by period.
- For drop-out bias, I would link company numbers to company size records, to see how many drop-outs fell below the size at which a company must report.
- For period effects, I would use every year from 2018 to 2024 in a model that allows for repeated measurements of each company, to see when the fall happened. Without a

comparison group it still could not say why. That needs a tool outside the brief's list of allowed tools.

- To weight by size, I would use the value totals that reports carry from 2025 (Reporting on Payment Practices and Performance (Amendment) Regulations 2024, reg. 2(4)(a)). They would describe the average pound paid, not the average company.

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